

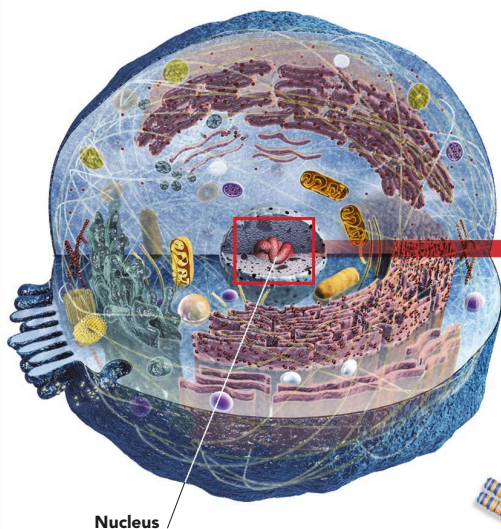
INHERITANCE

THE PASSING OF GENETIC INFORMATION FROM PARENT TO CHILD IS KNOWN AS INHERITANCE. THE INFORMATION IS CONTAINED IN CHEMICAL CODES CARRIED BY DEOXYRIBONUCLEIC ACID (DNA) IN THE SEX CELLS (EGGS AND SPERM).

INHERITANCE OF GENES

Everything that specifies a person is found in their genes. Each gene carries a “blueprint” to make a particular product, some of which affect appearance or biology – skin pigment, for instance. Other gene products combine

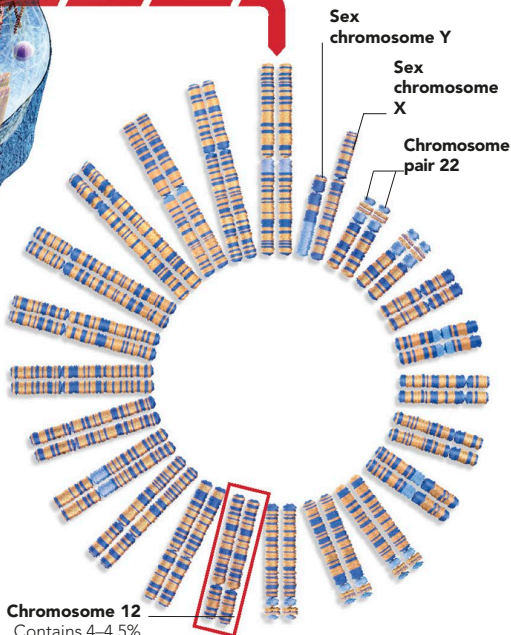
to produce a complex trait, such as athletic ability. Simple features controlled by single genes are inherited in predictable patterns (see pp.290–93). However, complex traits, such as height, are controlled by many genes.



Nucleus

CELL

The genetic information of DNA is based in the cell's nucleus. Almost every type of body cell carries DNA, but a few specialized types, such as red blood cells, lose it as they mature.



Sex chromosome Y

Sex chromosome X

Chromosome pair 22

Chromosome 12
Contains 4–4.5% of total genomic DNA; carries 1,000–1,300 genes

GENOME

The full complement of genetic instructions, known as the human genome, is carried by 23 pairs of chromosomes (see p.34). They represent about 20,000 protein-coding genes – a relatively low number for an organism as complex as the human being. The tiny fruit fly *Drosophila*, long used for genetic experiments, has 13,600 genes.